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Learning to Act with World Models by Valuing What Comes Next

WM-PPO: A World Model Predicts "What Comes Next"; PPO's Critic Values It

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Credit Assn. | World Models | PPO critic | ALFWorld / ScienceWorld

In a Nutshell

Valuing each candidate's next step makes PPO learn faster and better

World model predicts each candidate's next observation → critic estimates success probability of the predicted successor → **comparison = auxiliary policy advantage; mean = auxiliary critic target**, both weighted by a gate calibrated on real branch outcomes; no teacher, no privileged info, inference uses only the policy.

ALFWorld

63.5% → 87.5%

PPO-160 → WM-PPO (greedy, matched real env. steps)

ScienceWorld

39.3% → 52.9%

PPO-160 → WM-PPO (greedy, matched real env. steps)

vs. best PPO variant

+6.5 pts

ScienceWorld： 52.9% vs SO-PPO 46.5% （p=0.004）

Inference cost

+0 models

Deploys policy only; training GPU-hours ≈ 2.7–3.8× PPO-160

0/1 end reward hides the faulty step; dynamics give clues

Example: wash a plate and put it in the cabinet

Find the plate

↓

Pick up the plate

↓

Missed: wash it at the sink

↓

Put it in the cabinet

R = 0

Only one terminal signal

More observation text ≠ closer to success: opening a cabinet may list many irrelevant items;
a brief "heating done" may satisfy a key precondition.

World Models

Learn "what each action changes"

But predicted observations alone don't say if success is more likely

↓

PPO's critic

With R∈{0,1}, γ=1: $V^{\pi}(s) = Pr_{\pi}(\text{success} \mid s)$

But only applied to visited states and executed actions

↓

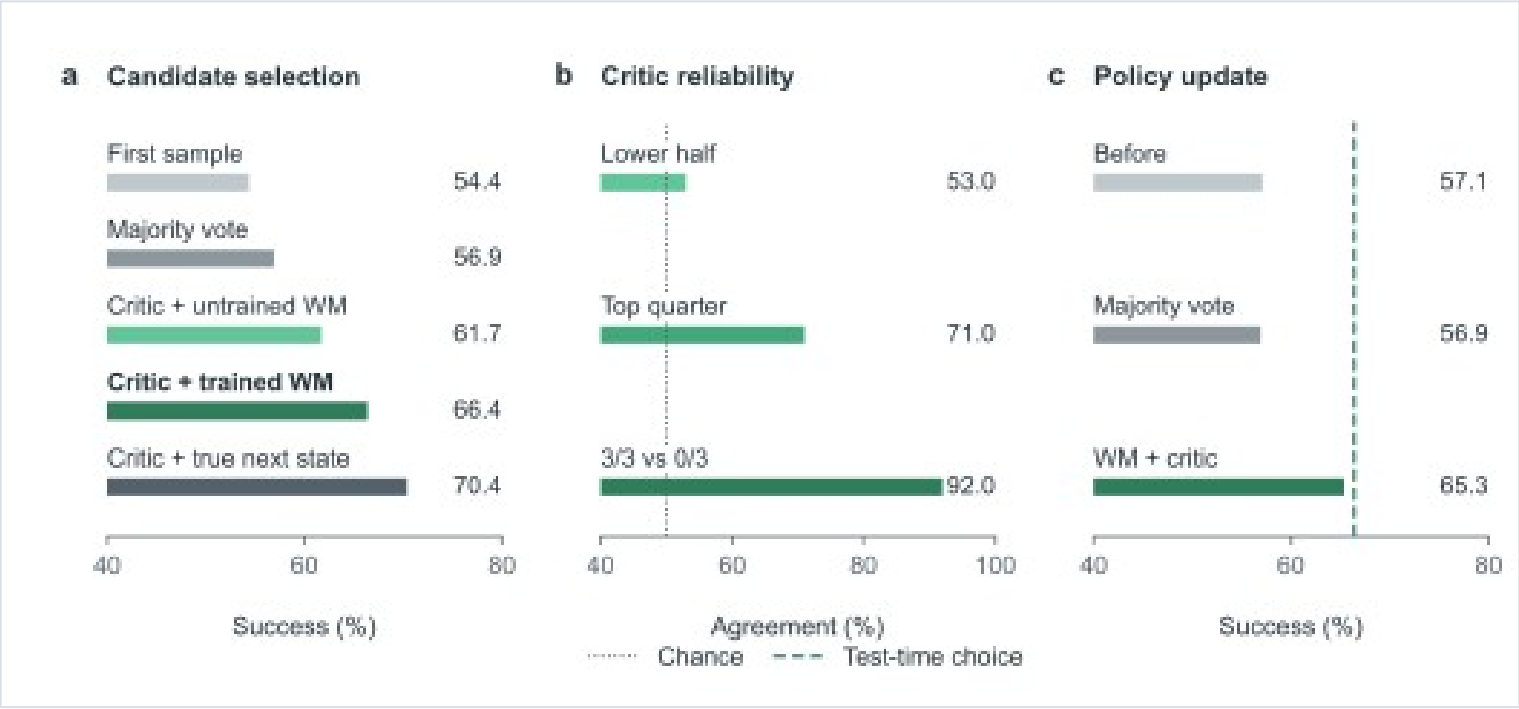
WM-PPO

Connect them: the critic values "predicted successors"

→ Candidates in the same state can be compared by consequences

Diagnostic Study

Critic on predicted successors: better actions, better policy



70.4%

Select actions by critic value of real successors (majority vote 56.9%, peek control 59.5%)

66.4%

Use trained world model's predicted successors (untrained world model: 61.7%)

57.1→65.3%

1 offline update with WM+critic labels, no task reward; majority-vote labels: 56.9%

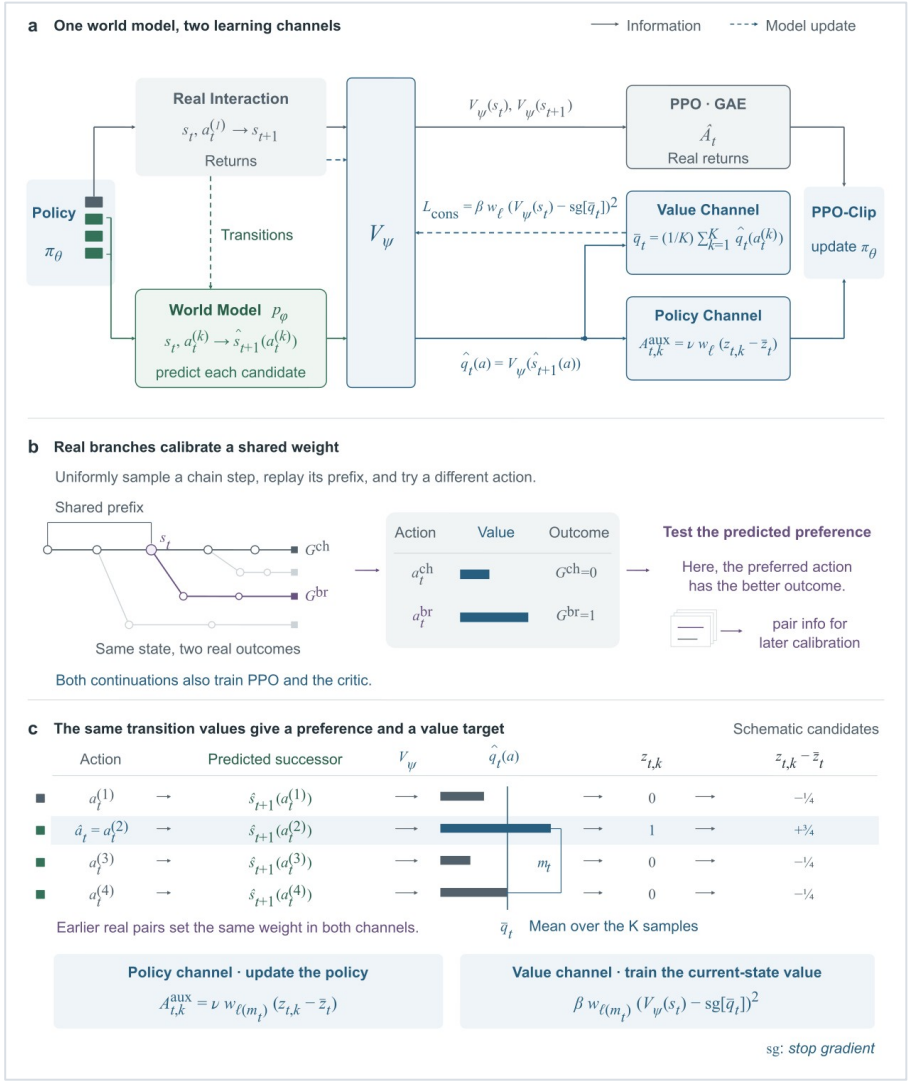
53%→71%

Critic ranking accuracy rises with value gap (bottom half → top quartile), hence calibration

Source: Fig. 1, § 2 (ALFWorld, PPO iter.-24 checkpoint, 274 episodes)

Method Overview

WM-PPO: one world model + one shared gate, two learning channels



① Sample candidates — Sample K=4 responses per chain state; execute the 1st (behavior policy unchanged)

② Predict outcomes — World model greedily predicts next obs. ô(a) for each distinct candidate, forming successor ŝ(a)

③ Value — q̂(a) = V_ψ(ŝ(a)); in ScienceWorld, a predicted end marker directly yields 1 or 0

④ Two channels — Comparison → auxiliary policy advantage; mean q̄ → auxiliary critic target

⑤ Gating — Main chain + 3 real branches; weights w_ℓ calibrated per value-gap bin, shared by both channels

Two Channels

One set of transition values gives policy preferences and critic targets

Policy channel

$$A_{aux} = v \cdot w_{\ell}(m) \cdot (z - \bar{z})$$

- z: whether a sample is the preferred action $\hat{a} = \text{argmax}\ \hat{q}$
- Zero-sum within group: probability mass shifts to the preferred action
- Learns only "which one", not "how much better"; v = 1, in the same clip objective as regular PPO samples

Value channel

$$L_{cons} = \beta \cdot w_{\ell}(m) \cdot (V_{\psi}(s) - sg[\bar{q}])^2$$

- $\bar{q}$ = mean $\hat{q}$ over K candidates, a one-step model estimate of $V^{\pi}(s)$
- $\beta = 0.5$; real-return regression remains the critic's main supervision
- More accurate critic → better GAE advantages in PPO's next iteration

Example (ScienceWorld, Fig. 19): task "make orange paint, then focus"; bowl already has red and yellow paint

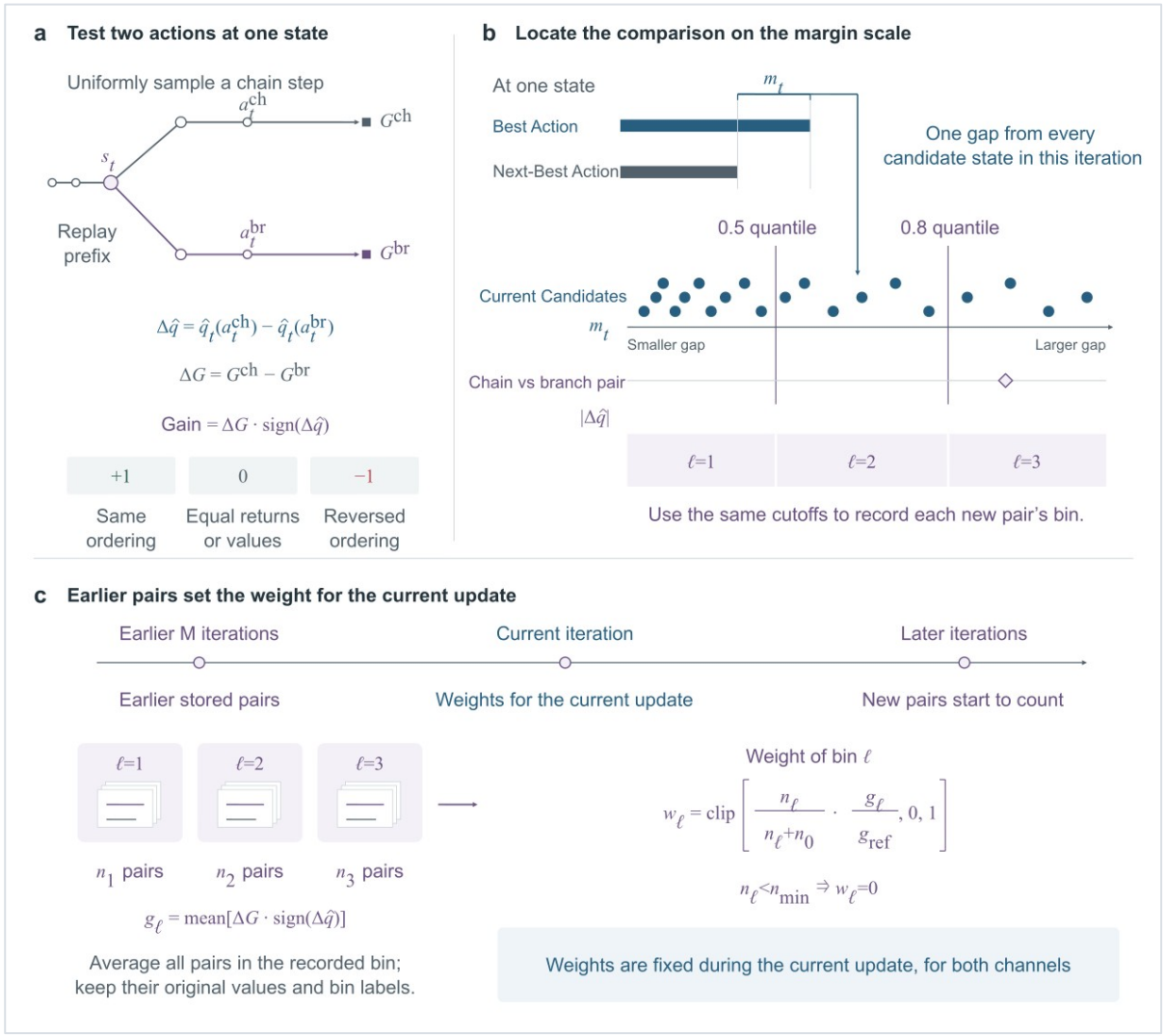
mix bowl	mix bowl	mix bowl	focus on orange paint
$A_{aux} = +1/4 \times w$	$A_{aux} = +1/4 \times w$	$A_{aux} = +1/4 \times w$	$A_{aux} = -3/4 \times w$

$\hat{q}(\text{mix}) - \hat{q}(\text{focus}) = +0.153$

World model predicts: mix → orange paint produced; focus → command rejected. Both real rollouts succeed, so terminal reward can't separate them, yet the preference still points to "mix first".

Key property (Eq. 7): $\hat{q}(a) - \hat{q}(a') \approx A(s,a) - A(s,a')$; the shared V(s) cancels, so comparisons don't depend on the critic's current-state value.

Real-branch calibration: trust only value gaps proven right



Gain = ΔG · sign(Δq̂)

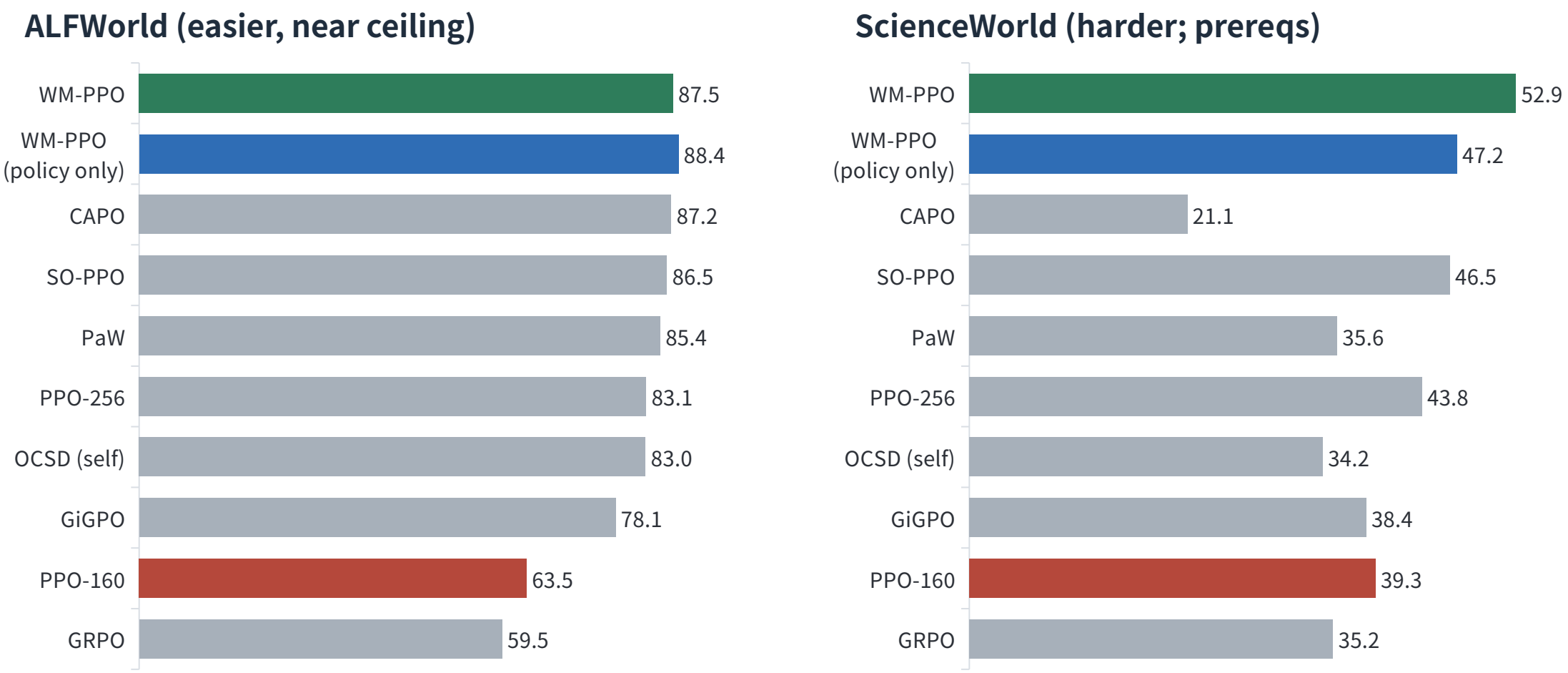
Same state, chain vs. branch: preference correct +1, wrong −1, same outcome 0

$w_\ell = \mathrm{clip}(n/(n+n_0) \cdot g_\ell/g_{ref}, 0, 1)$

If pair count n < n_min, the bin is closed (w = 0)

- 3 bins: 0.5 / 0.8 quantiles of this iteration's nonzero margins
- Pairs from past M = 4 iterations; $n_0 = 40$, g_ref = 0.05, n_min = 20
- Set once in an ALFWorld pilot, reused unchanged in both environments
- Gate closed = plain tree PPO; one weight acts on both channels

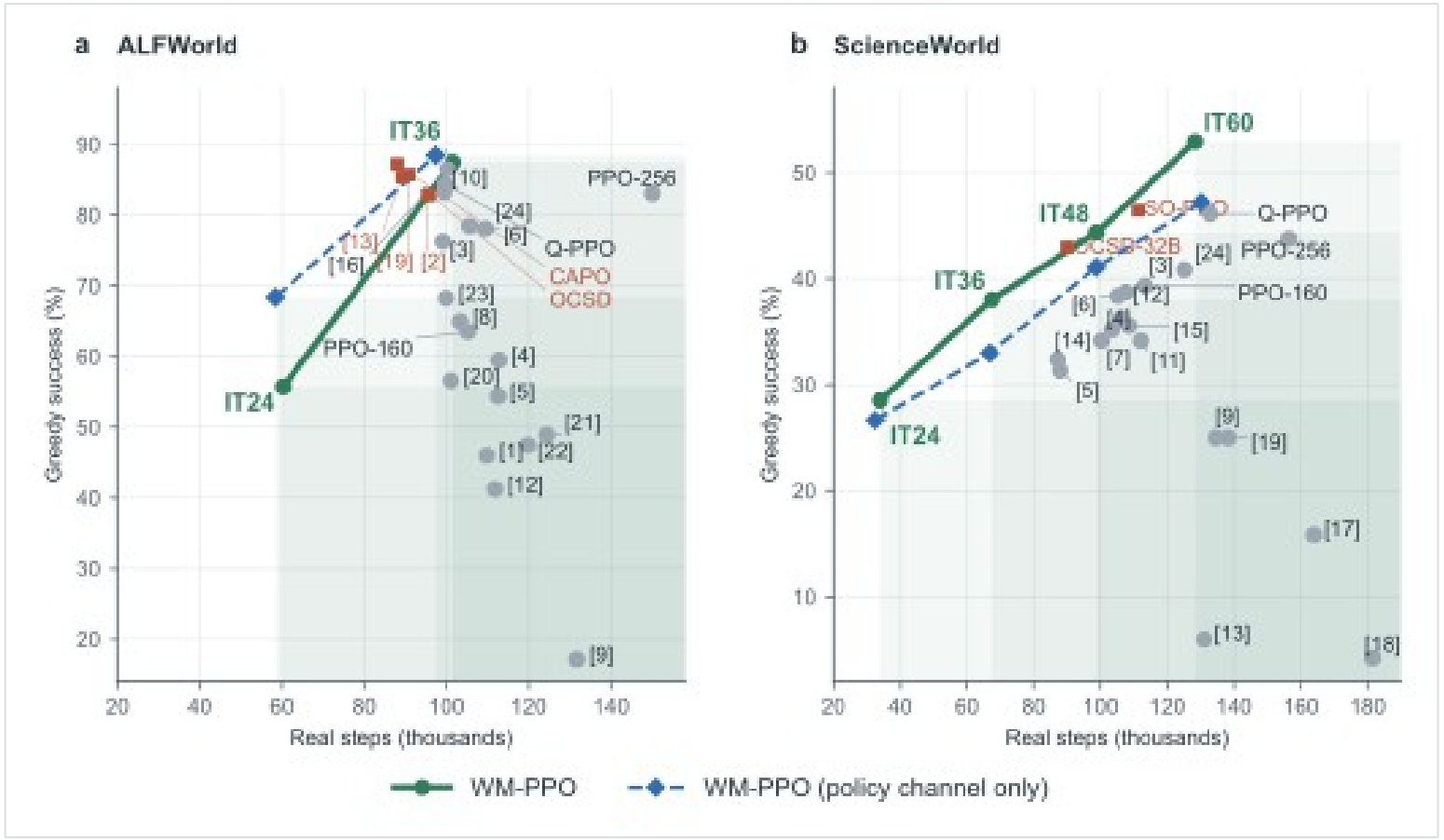
ScienceWorld: +6.5 pts over best baseline; ALFWorld: on par



Green: WM-PPO; blue: policy channel only; red: step-matched PPO-160. On ScienceWorld, CAPO diverged in all three settings; best pre-divergence checkpoint shown (21.1%). Every WM-PPO run beats every PPO-160 run.

Source: Table 2 (greedy success %, ALFWorld iter. 36 / ScienceWorld iter. 60)

Higher success at equal real steps; fewer steps to match



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Other methods in the dominated region on ALFWorld

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Other methods in the dominated region on ScienceWorld

69–97k vs 129–141k

ALFWorld to 80%: WM-PPO vs PPO-256 (PPO-160 not reached by iter. 36)

94–120k

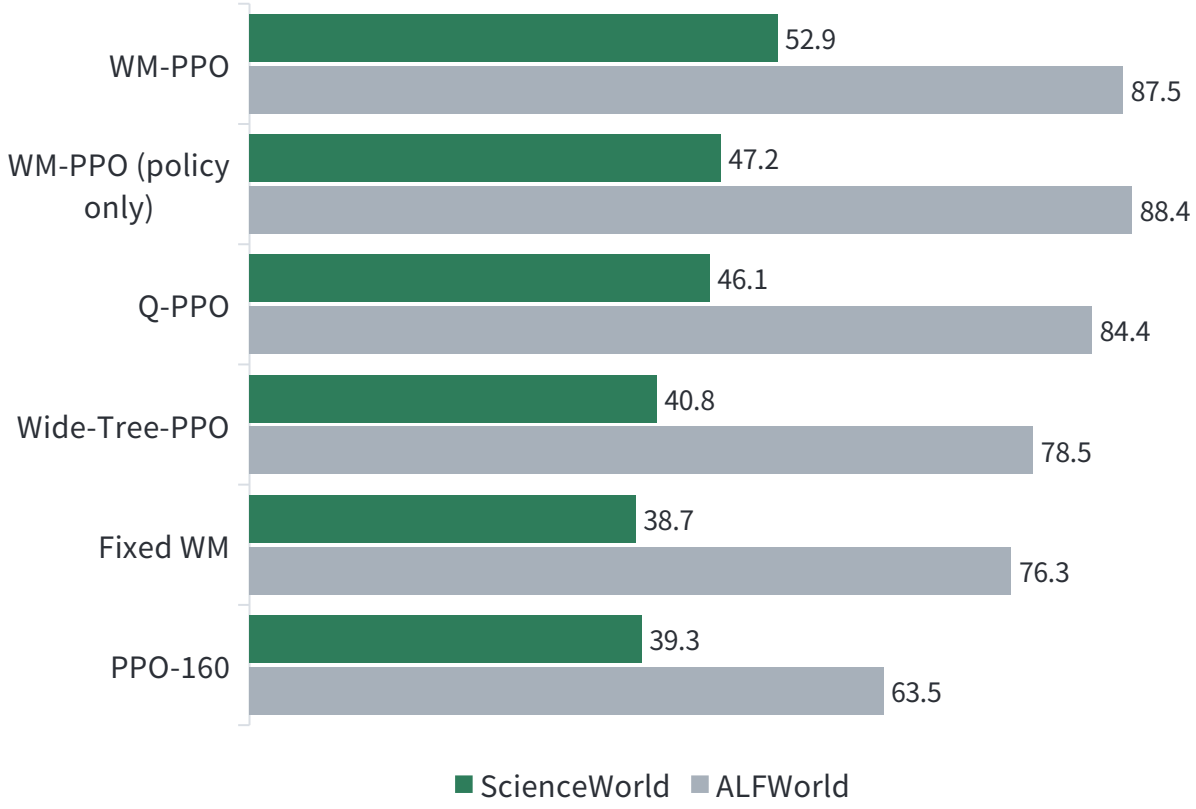
ScienceWorld to 45%: WM-PPO; PPO-160 not reached within 108–118k steps

Cost: per-iteration wall-clock 1.3–1.6×, GPU-hours 2.7–3.8× (training only)

Source: Fig. 4, Fig. 13 table (Table 14), Table 17

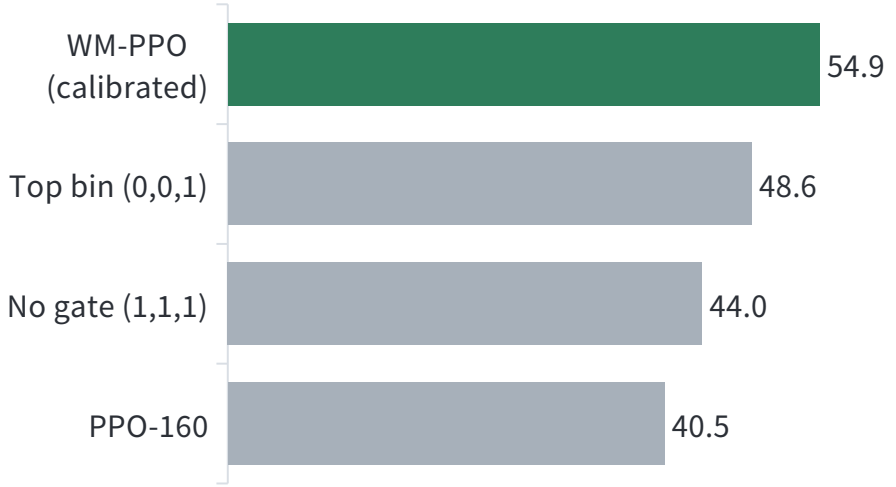
Ablations: no aux. updates, frozen WM, or no gate all hurt

Structural controls (greedy success %)



For Wide-Tree-PPO on ScienceWorld, at 82% of branch points all candidates get identical terminal returns; real outcomes can't separate them, but transition values still can.

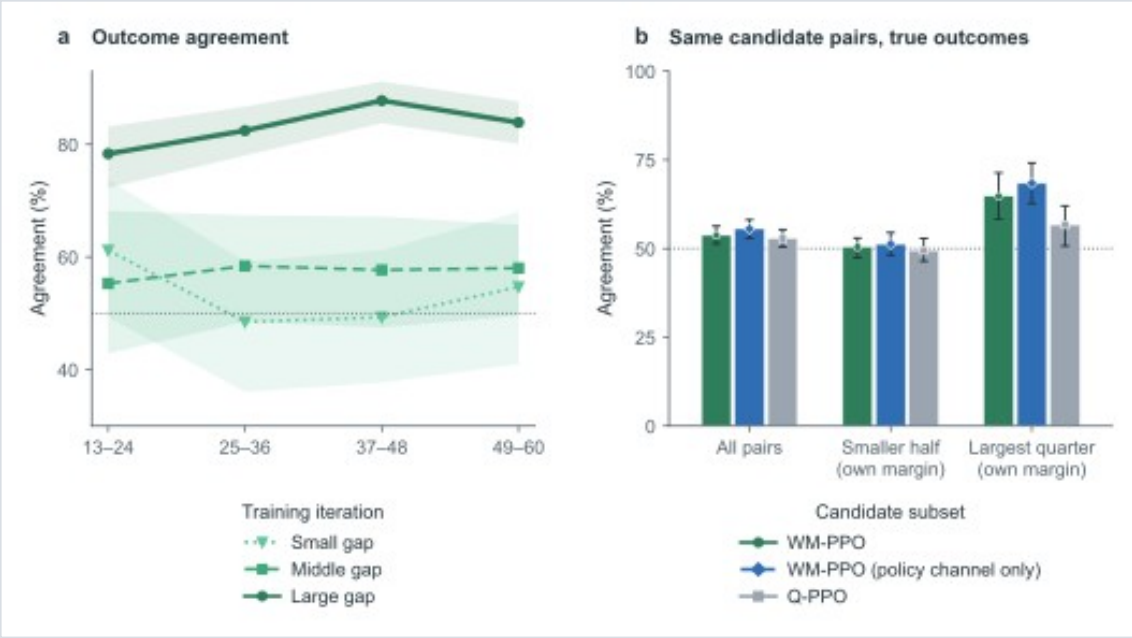
Gate ablation (ScienceWorld iter. 60, single seed)

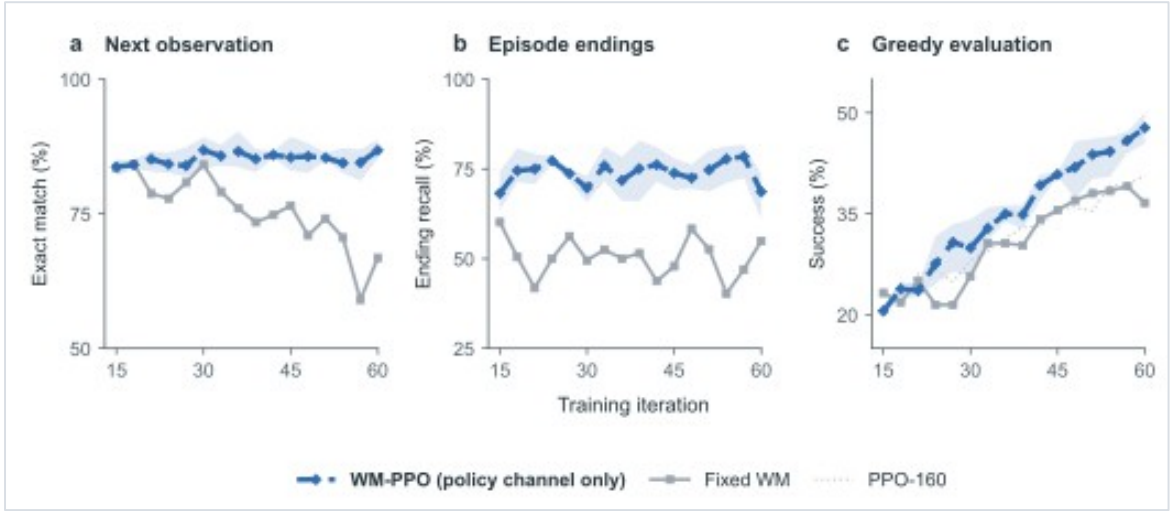


- **Wide-Tree:** same tree, aux. updates removed → −10.0 / −6.3 pts
- **Fixed WM:** world model not updated → ScienceWorld falls to PPO-160 level (38.7 vs 39.3)
- **Q-PPO:** directly learning Q values → worse than "predicted successor + V"
- **Gate:** all bins open injects near-random preferences; only the largest bin discards weak signals

Source: Table 2 (run means; Fixed WM single run), Table 21 (gate ablation, single seed)

Large-gap comparisons improve; online WM stays accurate





Largest-margin bin agreement: 78% → 88% (84% late)

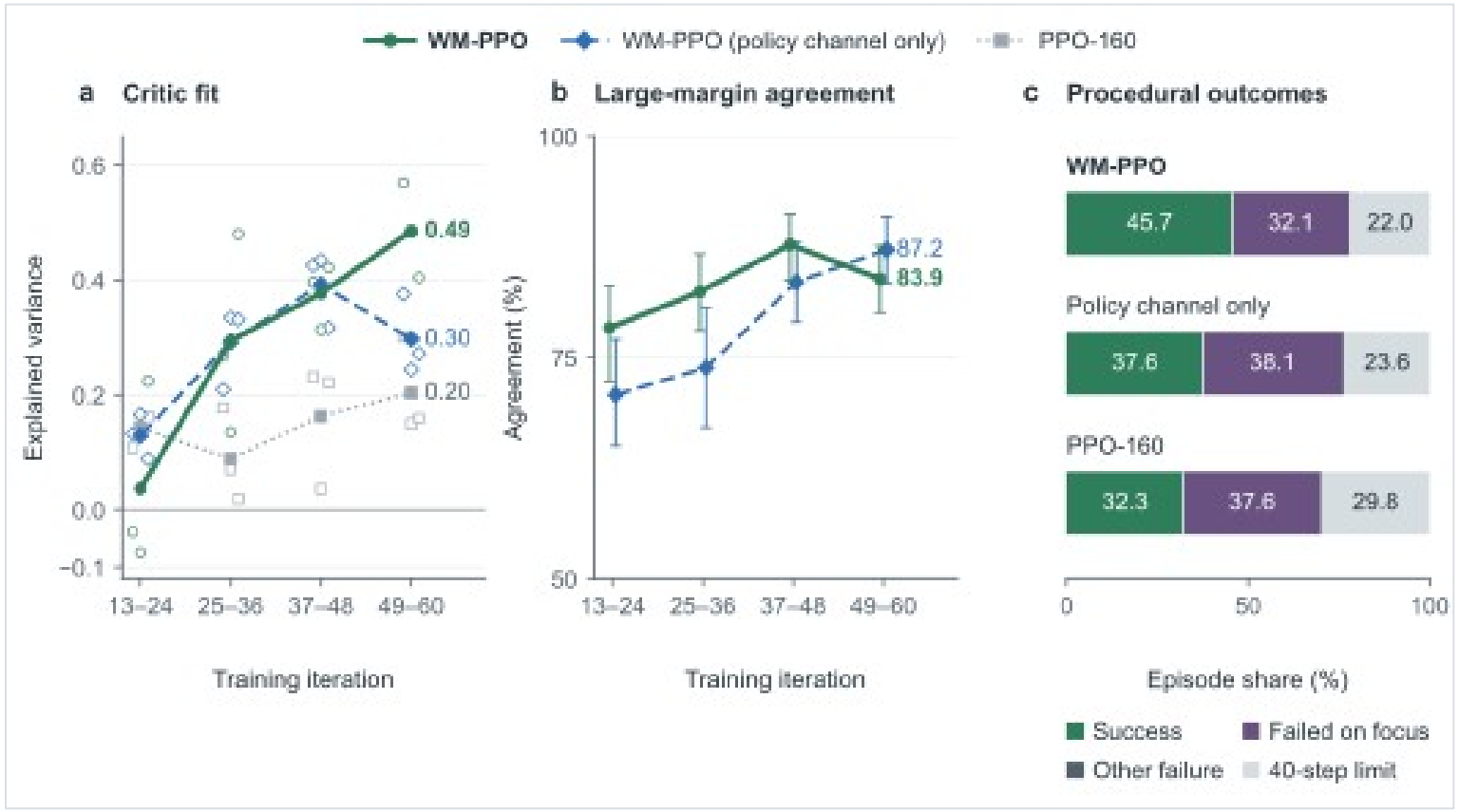
Small/medium bins stay at 50–60%, near random. In the largest bin: predicted successor + V agrees 85%, Q-PPO (learning Q directly) 76%. E.g., focus on air valued 0.03–0.34; all 24 real rollouts failed.

Online WM: exact prediction 84–86%; fixed WM: 81% → 68%

End recall ≈74% vs ≈50%. In the fixed-WM run, late largest-bin agreement is only 53–64%, with success on par with PPO-160.

Source: Fig. 5, Fig. 6, Tables 19/20/23 (ScienceWorld)

Value channel: better critic, earlier gate, ScienceWorld +5.8 pts



0.49 vs 0.30

Late critic explained variance (WM-PPO vs policy channel only; PPO-160: 0.20)

78–82% vs 71–74%

Early agreement of largest-margin comparisons

55% → 62%

Comparisons with both successors critic-valued (with predicted end: ≈95%)

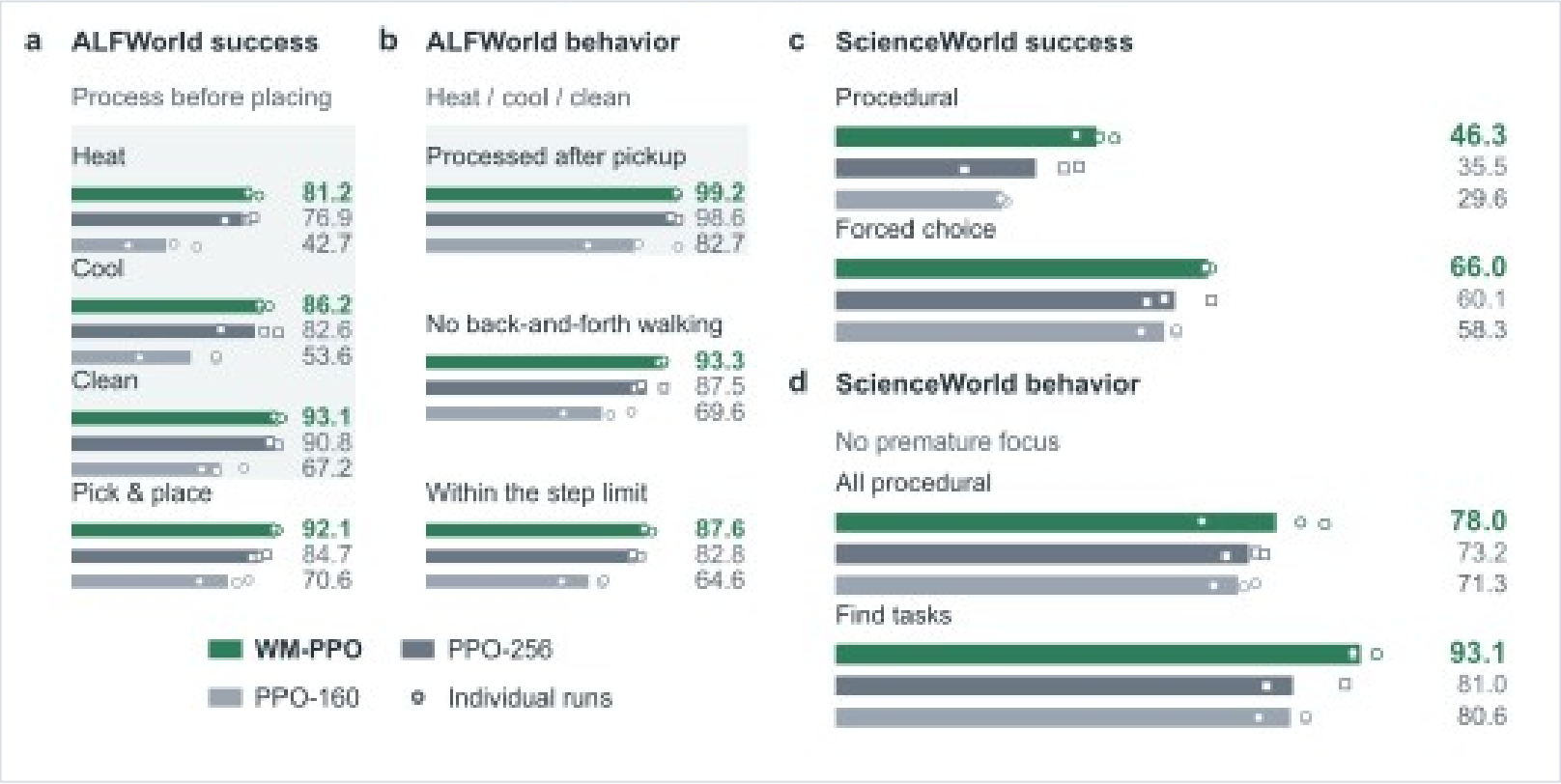
+5.8 pts

ScienceWorld greedy success: 52.9% vs 47.2%

Note: ALFWorld is near ceiling with few outcome-differing branch pairs, so the gate withdraws weight (largest bin closed at iter. 36); the value channel adds no gain (87.5% vs 88.4%, not significant).

Source: Fig. 7, Fig. 8, Table 20 (ScienceWorld)

Learned: meet preconditions before placing or answering



81.2% vs 42.7%

ALFWorld Heat tasks: WM-PPO vs PPO-160

41% → 17%

Wrong order after pickup: "place first, process later"

46.3% vs 29.6%

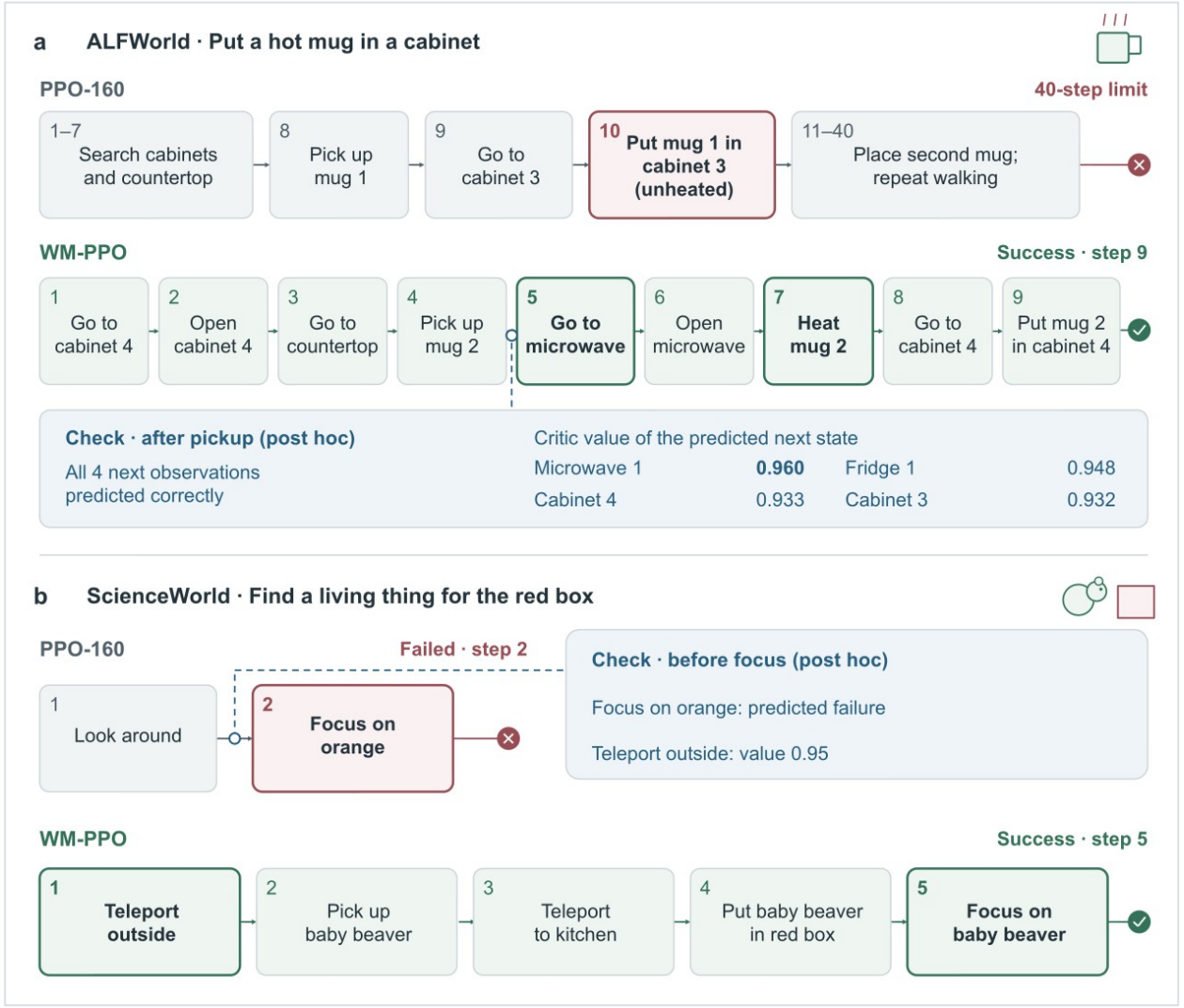
ScienceWorld procedural task success

28.7% → 22.0%

Premature focus (answering before preconditions); find tasks: 19.4% → 6.9%

Smaller gains on forced-choice tasks: correctness shows only at the end.

Two cases: PPO-160 skips prereqs or rushes; WM-PPO sets them up



ALFWorld: put a hot mug in the cabinet

PPO-160: puts it in the cabinet unheated → hits 40-step limit

WM-PPO: heats it in the microwave first, succeeds at step 9; post-hoc critic: microwave highest at 0.960

ScienceWorld: find a living thing

PPO-160: focuses on orange at step 2 → fails

World model predicts this step fails; teleport outside valued 0.95. WM-PPO succeeds at step 5

Source: Fig. 11 (greedy episodes of final policies; Check = post-hoc check)

Costs/limits: training-only overhead; needs replayable envs

Training cost (vs PPO-160)

1.3–1.6×	Per-iteration wall-clock (ALFWorld 40 vs 32 min; ScienceWorld 55 vs 34 min)
2.7–3.8×	GPU-hours (upper bound: 7 GPUs, shared server fully counted)
≈ PPO-256	Total env. calls incl. prefix replay
0	Extra models at inference: deploy policy only

Limitations (acknowledged by authors)

- Only one 4B base model (Qwen3-4B) and two text environments (plus Sokoban); ⩽ 40 steps per episode
- Branching and gating require deterministic replay or saved env. states
- Branch actions are sampled under conditions differing from the chain; updates are uncorrected
- Single greedy prediction, no uncertainty; exact rate for observations revealing new objects only 26%
- Long-horizon effects rely entirely on the critic; prediction errors can bias auxiliary signals
- Many small-margin comparisons get near-zero weight and go unused

Key Takeaways

1

Value predicted consequences

The world model says "what will happen"; the critic says "what it means for success"; both learn only from own interaction, and deployment uses only the policy.

2

Calibration matters more than quantity

Critic comparisons are near random overall; only large value gaps are reliable. A shared gate calibrated online on real branches is key to stability.

3

Less interaction, better behavior

ALFWorld 63.5→87.5%, ScienceWorld 39.3→52.9%; the policy learns to satisfy preconditions before placing or answering.

Best suited for: sparse-reward multi-step tasks with slow/costly real interaction and replayable envs; extending to stochastic or physical settings first requires solving state replay and prediction uncertainty.